

ORIGINAL ARTICLE



Role of mTOR pathway in modulation of radiation induced bystander effects

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ABSTRACT

Purpose: Radiation-induced bystander effect (RIBE) is considered as an important consequence of radiation exposure. Based on the type of effect induced, it has important implications in radiation therapy. mTOR pathway, a key regulator of cell survival, plays an important role in radiation-induced damages. However, the role of mTOR signaling in the modulation of RIBE is still unclear. We evaluated the role of mTOR pathway in RIBE and its relationship with the radiation response of target cells.

Materials and methods: Direct and bystander effects were evaluated by using clonogenic and MTT assay in five different cell lines. Expression of mTOR pathway proteins in directly targeted and bystander cells was studied using western blotting.

Results: Among five different cell lines naïve HT1080 and A549 cells exhibited proliferative bystander effect induced by conditioned media and irradiated conditioned media, while no effect was observed in other cell lines. Everolimus significantly abolished the proliferative bystander effect induced in naïve cells.

Conclusions: These results suggested that the mTOR pathway plays an important role in RIBEs. These effects are cell type-specific and depending on the radiosensitivity of the target cells, therapeutic benefits of radiation may be modulated by treatment with mTOR inhibitors.

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Introduction

The non-targeted effects are reported in cells/tissues/organs which are not directly traversed by radiation but show characteristics similar to irradiated cells. These effects have been termed radiation-induced bystander effects (RIBEs). RIBEs have been observed in cells that are either in the neighborhood or are distantly located from irradiated cells. These effects can expand the targeted volume of radiation in the form of genomic instability (Watson et al. 2000), mutations (Wu et al. 1999), neoplastic transformations (Lewis et al. 2001), changes in gene expression (Sokolov and Neumann 2018), etc. in non-targeted cells.

Based on the responses of cells to therapeutic radiation, bystander effects have been classified into two types: Type I – Inhibitory bystander effect (*decreased* survival of bystander cells) and Type II – Proliferative bystander effect (*increased* survival of bystander cells) (Taychisiraprapakul et al. 2012). Proliferative bystander effect can be characterized by an increase in cell survival greater than 1.0 that is generally associated with various processes like reduced cell death, enhanced genomic instability, radioresistance, cell differentiation, cell cycle progression, etc. These proliferative bystander effects correspond with hallmarks of cancer development and hence can enhance the risk of development of second cancers (Baskar 2010; Heeran et al. 2019). Besides

proliferative effect, the inhibitory bystander effect is characterized by decreased cell survival which can lead to reduced cell proliferation or increased probability of tumor regression by destroying damaged cells or killing cancer cells in distant locations after radiotherapy (Wersäll et al. 2006).

A number of factors can influence the bystander responses, for example, cell type used, type of radiation used, donor cell density, radiation dose, time of media transfer after irradiation, etc. Multiple signaling pathways and signaling molecules have been reported to be involved in the induction and modulation of bystander effects (Verma and Tikku 2017). In the last few decades signaling pathways belonging to the MAPK super-family have been explored which are responsible for cell growth, proliferation, differentiation, apoptosis or even senescence (Cargnello and Roux 2011; Anerillas et al. 2020; Yue and López 2020). They have been reported to play a role in RIBE (Asur et al. 2010; Faqihi et al. 2015). Studies on conditioned media harvested from irradiated cells suggest that changes in several cytokine levels and medium soluble factors are responsible for the induction of different bystander effects (Siva et al. 2015). However, the exact factors/signals that guide the induction of proliferative (could lead to cancer development) or inhibitory (could lead to tumor regression) bystander effect is yet to be understood. Therefore, the major challenge/gap in this field is to explore the signal molecules/signaling

pathway(s) responsible for the induction of different types of bystander effects.

mTOR (mammalian Target of Rapamycin) signaling is linked to the regulations of multiple stress-mediated as well as normal cellular events like cell growth, metabolism, proliferation, protein synthesis, senescence, autophagy, etc. (Weichhart 2018). It is also reported to be involved in increased cell survival and resistance in response to radiation (Kim et al. 2010; Zheng et al. 2016). Since the mTOR pathway plays an important role against radiation-induced stress, we have tried to understand its role in the induction/modulation of RIBEs.

Materials and methods

Materials

Roosevelt Park Memorial Institute (RPMI)-1640 medium, fetal bovine serum (FBS) and antibiotic solution were purchased from Hi-Media Laboratories Pvt. Ltd. Mumbai, India. Everolimus (Ev), 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), bovine serum albumin (BSA), Triton-X 100 and propidium iodide (PI) were bought from Sigma (Sigma-Aldrich, St. Louis, USA). Dimethylsulfoxide (DMSO), crystal violet, glacial acetic acid were obtained from Merck (Germany). Enhanced chemiluminescence (ECL) and skimmed milk were purchased from Bio-Rad Laboratories (Berkeley, California). Primary antibodies [procaspase-3, p-Akt, p-mTOR, p-4EBP1, p-P70S6K, GAPDH] and horseradish peroxidase (HRP)-conjugated secondary antibodies were purchased from Cell Signaling Technology (Beverly, MA, USA). All other reagents/chemicals used in different experiments were of analytical grade.

Cell culture

A normal cell line: L132 (human alveolar epithelial cells) and four cancer cell lines: HT1080 (human epithelial fibrosarcoma), A549 (human alveolar epithelial adenocarcinoma), MCF-7 (human epithelial breast adenocarcinoma) and HT29 (human colorectal adenocarcinoma) were used for the present study. These cells were obtained from National Center for Cell Sciences, Pune, Maharashtra, India and maintained as monolayer cultures (2D) in the cell culture facility. Cells were grown in tissue culture flasks (Corning) using RPMI-1640 medium supplemented with 10% FBS and 1% antibiotics at 37°C with 5% CO₂. Cells were maintained in exponentially growing culture conditions and passaged twice a week.

Radiation treatment

Cells were irradiated with Co⁶⁰ γ -rays in a gamma chamber (240 TBq Co⁶⁰ Model 4000 A) obtained from the Isotope Division of Bhabha Atomic Research Center (BARC), Mumbai, India. Cells were exposed to different doses of radiation (0–8 Gy) and the dose rate was 2.11 Gy/min. All irradiations were carried out at room temperature.

Media transfer method and treatments used for bystander effects

For the media transfer method (Figure 1(A,B)), 3×10^6 cells were seeded in 100 mm Petri-plates and allowed to attach. After overnight incubation, cells were treated with different doses of radiation or different doses of everolimus (Ev) for 1 h or Ev + radiation (1 h Ev treatment followed by media replacement and radiation exposure). Sham-treated samples were used as controls. After different treatments, cells were incubated in fresh media for 3 h at 37°C. 3 h after incubation, media was harvested and centrifuged at $300 \times g$ for 5 min. The supernatant was collected and filtered through a 0.22 μ m syringe filter. Different conditioned media were collected as follows

- **Condition Media (CM)** – Sham irradiated cells were incubated in normal culture media and this media was collected after 3 h
- **Irradiated Conditioned Media (ICM)** – Cells were irradiated with different doses of radiation and media was collected from these cells 3 h post-irradiation
- **Everolimus treated Conditioned Media (Ev CM)** – Cells were treated with Ev for 1 h, drug-containing media was removed and fresh media was added. The media was then collected after 3 h
- **Everolimus treated Irradiated Conditioned Media (Ev ICM)** – Cells were treated with Ev for 1 h, drug-containing media was removed and fresh media was added. Cells were then irradiated with different doses of radiation and media was collected 3 h post-irradiation

The conditioned media thus were immediately transferred to naïve cells for the study of bystander effects.

Clonogenic assay

Approximately 1×10^5 cells were seeded in 35 mm Petri-plates and incubated overnight. The cells were then treated with different doses of radiation (0–8 Gy) or Ev (0–100 nM) for 1 h. For Ev + radiation treatment, Ev treatment was given for 1 h followed by radiation exposure. Depending on the treatment, 300–3000 cells were seeded in 60 mm Petri-plates with 5 ml of complete media and incubated for 7–12 days at 37°C for cells to grow and form colonies.

For bystander effect, 5 ml conditioned media was harvested from cells 3 h post-treatment. 300–500 naïve cells were grown in this media at 37°C for 7–12 days. After required incubation, colonies were washed with PBS and fixed in 3.7% formaldehyde for 30 min. Fixed cells were then stained with crystal violet solution for 20 min. Excess stain was drained off and plates were rinsed with PBS. The colonies with >50 cells were counted by ImageJ software. The survival fraction was calculated as follows:

Plating efficiency of treatment group (targeted group or bystander group) / Plating efficiency of control group

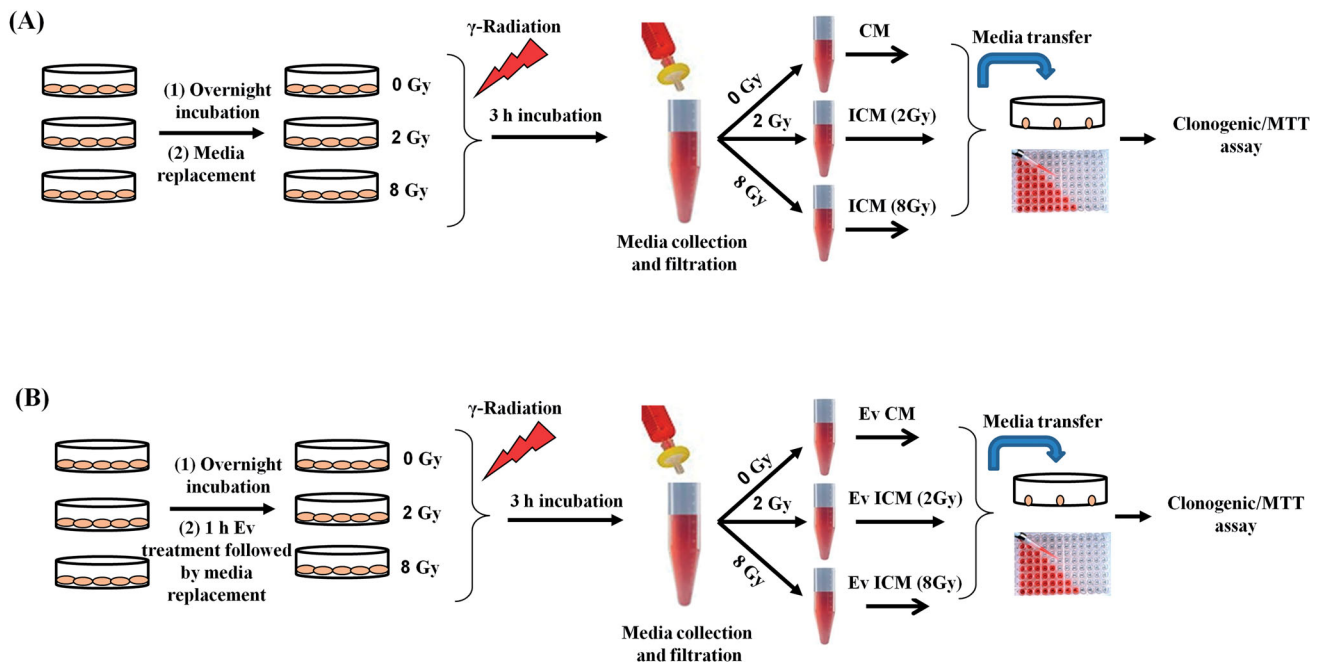


Figure 1. Schematic diagram showing media transfer experiments for induction of bystander effects using (A) Only radiation (B) Ev or Ev + radiation. CM: Conditioned media from sham irradiated cells, ICM: Conditioned media from irradiated cells, Ev CM: Conditioned media from everolimus treated cells, Ev ICM: Conditioned media from everolimus + radiation treated cells.

Cell proliferation assay

To determine cell viability MTT assay was performed. For bystander effect, 1000 cells/well were seeded with 2 ml conditioned media in 12 well plates. After 96 h incubation, 200 μ l MTT (2 mg/ml) was added to each well. Media with MTT was removed after 4 h incubation and 500 μ l of DMSO was added to each well. The plates were kept at room temperature for another 5–10 min till the purple color developed. Absorbance was measured at 595 nm using Bio-Red microplate reader.

Immunoblotting

For direct effect, cells were seeded in 100 mm Petri-plate. After overnight incubation, cells were treated with or without Ev and/or radiation and incubated for 3 h. For bystander effect, naïve cells were incubated for 72 h in conditioned media harvested from 3 h post-treated cells. After required incubation, cells were lysed and scraped in RIPA buffer at 4 °C. Protein estimation was done by the Bradford method. 50 μ g of proteins from each sample was run on SDS-PAGE and then transferred to nitrocellulose membrane. Blocking of the membrane was done for 1 h by using 5% BSA in tris-buffered saline + 0.5% tween 20 (TBST) and probed using primary antibodies. Primary antibodies were detected using HRP-labelled secondary antibodies, which in turn were visualized by ECL solution.

Statistical analysis

All statistical analysis was done by using Prism 6, GraphPad Software, Inc. Values are expressed as the mean $\pm$ SD of three independent experiments. The significance of the

difference between the control and each test condition was analyzed by one-way and two-way ANOVA and $p < .05$ was considered statistically significant.

Results

Radiation-induced direct effect

The direct effect of different doses of γ -irradiation (0–8 Gy) on various cell lines was assessed by clonogenic assay. A dose-dependent decrease in cell survival was observed in all cell lines after radiation exposure. HT1080, A549, and MCF-7 cells were relatively more radiosensitive than L132 and HT29 cells (Figure 2(A)). The LD₅₀ value ranged from 1.9 Gy for HT1080 to 3.95 Gy for HT29 cells. These data suggested that among all cell lines studied, HT1080 cells show maximum radio sensitivity while HT29 cells show maximum radioresistance (Figure 2(A)).

Radiation induced bystander effect

To investigate the bystander effect, naïve cells were treated with CM or ICM harvested from different cell lines (as shown in Figure 1(A)) and incubated for 7–12 days at 37 °C to grow and form colonies (Figure 2(B) and Supplementary Figure 1(A)). The CM from sham irradiated HT1080 and A549 cells significantly induced cell proliferative bystander effect in autologous cell lines while CM from MCF-7, L132 and HT29 cells did not induce any effect (Figure 2(C)). Further, ICM (8 Gy) from HT1080 cells showed a significantly increased cell survival as compared to CM from sham irradiated cells. Other cell lines did not show a significant proliferative bystander effect after treatment with ICM as compared to CM (Figure 2(E)). Further, the MTT assay also

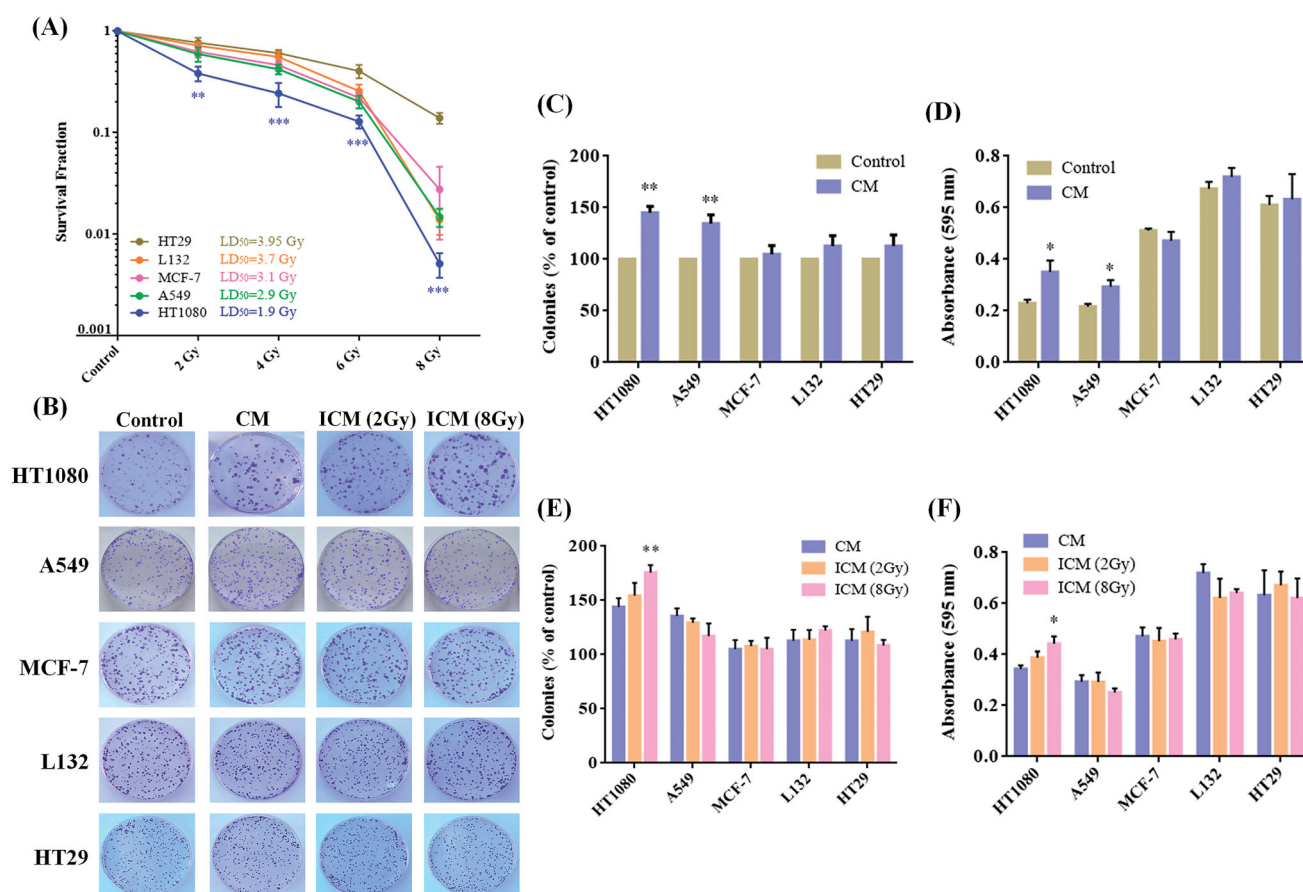


Figure 2. Direct and bystander effect of different doses of γ -irradiation on cell survival. (A) Direct effect of radiation on cell survival of HT1080, A549, MCF-7, L132 and HT29 cells using clonogenic assay. Mark increased radiosensitivity of HT1080 with respect to HT29 cells. (B) Representative cell culture petri-dishes showing cell survival of different cell lines after CM and ICM treatment. (C, D) Clonogenic and MTT assay showing effect of CM on cell survival with respect to control. (E, F) Clonogenic and MTT assay comparing the effect of ICM on cell survival with respect to CM. Data are means $\pm$ SD from three independent experiments (* $p < .05$, ** $p < .01$, *** $p < .001$).

confirmed the bystander effect induced by CM or ICM from different cell lines (Figure 2(D,F)).

Everolimus induced direct effect

Everolimus (Ev), an analog of rapamycin, was used to check the effect of inhibition of mTOR signaling on the proliferation of different cell lines. A dose-dependent decreased cell survival was observed in HT1080, A549 and MCF-7 cells, however, no significant effect was observed in the case of L132 and HT29 cells (Figure 3(A)). The IC₅₀ values for HT1080, A549 and MCF-7 cells were found to be less than 30 nM indicating an intrinsic sensitivity to Ev while L132 and HT29 cells were resistant to Ev.

Everolimus induced bystander effect

To check the bystander effect induced by Ev, conditioned media from Ev treated cells (Ev CM) was harvested from different cell lines (as shown in Figure 1(B)). The Ev CM from HT1080 cells significantly reduced the cell viability of autologous naïve cells in a dose-dependent manner while Ev CM from A549 and MCF-7 cells showed toxicity at higher doses only (Figure 3(B)). Ev CM from L132 and HT29 cells

did not show any toxic bystander effect in autologous cell lines (Figure 3(B)). These data suggested that among different cell lines, naïve HT1080 cells are most susceptible to Ev CM-induced toxic bystander effect.

Ev + radiation induced direct effect

To check the effect of mTOR inhibition on radioresponse of different cell lines, 100 nM Ev treatment was given for 1 h prior to irradiation. The clonogenic assay showed that Ev treatment increased the radiosensitivity of HT1080, A549 and MCF-7 cells (Figure 4(A)). Exposure to 2 Gy radiation reduced the cell viability of HT1080 cells to 40%, while Ev + 2 Gy radiation treatment reduced it further up to 90%. However, no significant change in cell viability or radioresponse was observed in Ev + radiation treated L132 and HT29 cells as compared to only irradiation groups (Figure 4(A)).

Ev + radiation-induced bystander effect

The effect of conditioned media from Ev + radiation treated cells [Ev ICM (2 Gy) and Ev ICM (8 Gy)] was analyzed in different cell lines by using clonogenic and MTT assay. Cell

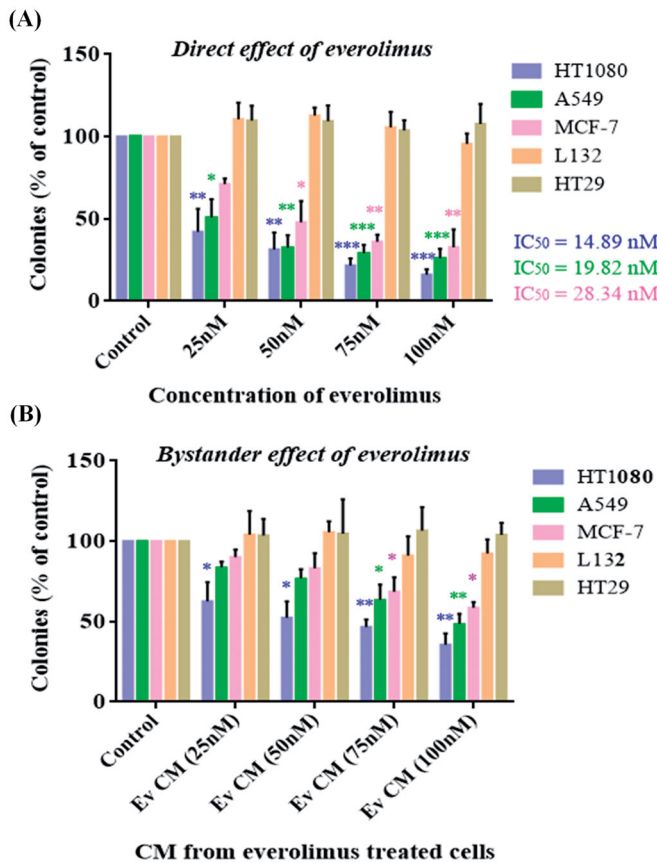


Figure 3. Direct and bystander effect of Ev on cell survival. (A) Direct effect of different doses (0–100 nM) of Ev on cell survival of various cell lines. (B) Effect of Ev CM from 0 to 100 nM Ev treated cells on autologous cell lines. Data are means $\pm$ SD from three independent experiments, (* p < .05, ** p < .01, *** p < .001) with respect to un-treated control.

viability was significantly reduced in bystander HT1080, A549 and MCF-7 cells while L132 and HT29 cells did not show any effect (Figure 4(B) and Supplementary Figure 1(B)). In naïve HT1080 cells, Ev CM treatment reduced the cell survival to 60% while Ev ICM (8 Gy) reduced it further to 90% (Figure 4(C)). MTT assay also confirmed the toxic effect of Ev ICM in three cell lines only (Figure 4(D)). These results suggested that among cell lines studied, Ev ICM showed a maximum toxic effect in naïve HT1080 cells.

The results obtained from the studies done so far suggest that among five different cell lines, HT1080 cells show maximum RIBE while L132 (normal lung) cells do not show any bystander effects (Supplementary Figure 3(A,C)). Thus, to explore the modulations induced by different conditioned media on bystander HT1080 and L132 cells, we studied the effect on cell density and cell morphology. A significantly increased cell density was observed in CM/ICM treated HT1080 cells while Ev CM/Ev ICM treated cells showed reduced cell density (Figure 5(A)). Cell morphology of bystander HT1080 cells also changed after Ev CM/Ev ICM treatment. Ev CM treated cells became elongated while Ev ICM treated cells showed small and rounded cell morphology. In contrast to the results obtained from bystander HT1080 cells, no significant change in cell density and cell morphology was observed in bystander L132 cells (Figure 5(B)).

Effects of different treatments on mTOR pathway proteins

Results from protein expression studies using western blotting showed that the proteins of the mTOR pathway expressed constitutively in all cell lines (Figure 6(A)). Radiation exposure significantly elevated the expression of phosphorylated Akt (Ser⁴⁷³), mTOR (Ser²⁴⁴⁸) and p70S6K (Thr³⁸⁹) proteins by 19-, 2- and 2.5-folds, respectively in HT1080 cells (Figure 6(A) and Supplementary Figure 2(A)) while A549 cells showed a non-significant increase of these proteins (Figure 6(A) and Supplementary Figure 2(B)).

Ev treatment significantly increased the level of p-Akt in all cell lines while no significant modulation in the expression of procaspase-3 was found in any of the tested cell lines (Figure 6(A)). As MCF-7 cells are caspase-3 deficient, no expression of procaspase-3 was expected hence no band was observed. Ev treatment reduced the expression of phosphorylated mTOR (Ser²⁴⁴⁸) and P70S6K (Thr³⁸⁹) proteins in HT1080, A549 and MCF-7 cells (Figure 6(A)). The expression of p-4EBP1 (Thr^{37/46}) was decreased only in HT1080 cells. In L132 and HT29 cells, no marked change in the expression of p-mTOR and p-4EBP1 proteins was observed while P70S6K was significantly reduced (Figure 6(A)). These observations suggested a constitutive activation of mTOR signaling in L132 and HT29 cells even after Ev treatment. This can be directly correlated with radioresistance in these cell lines.

Ev + radiation treatment significantly increased the expression of phosphorylated Akt in all cell lines while no significant change in the expression of procaspase-3 was observed (Figure 6(A) and Supplementary Figure 2). A reduced expression of phosphorylated mTOR, 4EBP1 and P70S6K proteins was detected in HT1080, A549 and MCF-7 cells. No marked change in the p-mTOR and p-4EBP1 proteins was observed in L132 and HT29 cells, however, the expression of p-P70S6K was decreased (Figure 6(A)). These results suggested that 100 nM Ev treatment either alone or in combination with radiation was not sufficient to induce cell death in all cell lines tested within 3 h of treatment, but is enough to inhibit the mTOR pathway in three cell lines.

Protein expression data also suggested that the mTOR pathway was significantly activated in HT1080 cells after radiation exposure; however, Ev treatment blocked its activation. In contrast, no significant effect of these treatments was observed in L132 and HT29 cells.

Further, HT1080 and L132 cells were selected to elucidate the effect of CM, ICM, Ev CM and Ev ICM on mTOR pathway in bystander cells. Western blotting results showed significant activation of Akt (3.8 folds), mTOR (1.6 folds), 4EBP1 (2 folds) and P70S6K (1.6 folds) proteins in bystander HT1080 cells after treatment with ICM for 72 h while no marked change in the expression of procaspase-3 was observed (Figure 6(B) and Supplementary Figure 3(B)). Ev ICM treated HT1080 cells showed significantly decreased expression of almost all the proteins. However, in the case of L132 cells, no marked change in the expression of procaspase-3 and activated mTOR pathway proteins (i.e. p-Akt, p-mTOR, p-P70S6K and p-4EBP1) was observed after

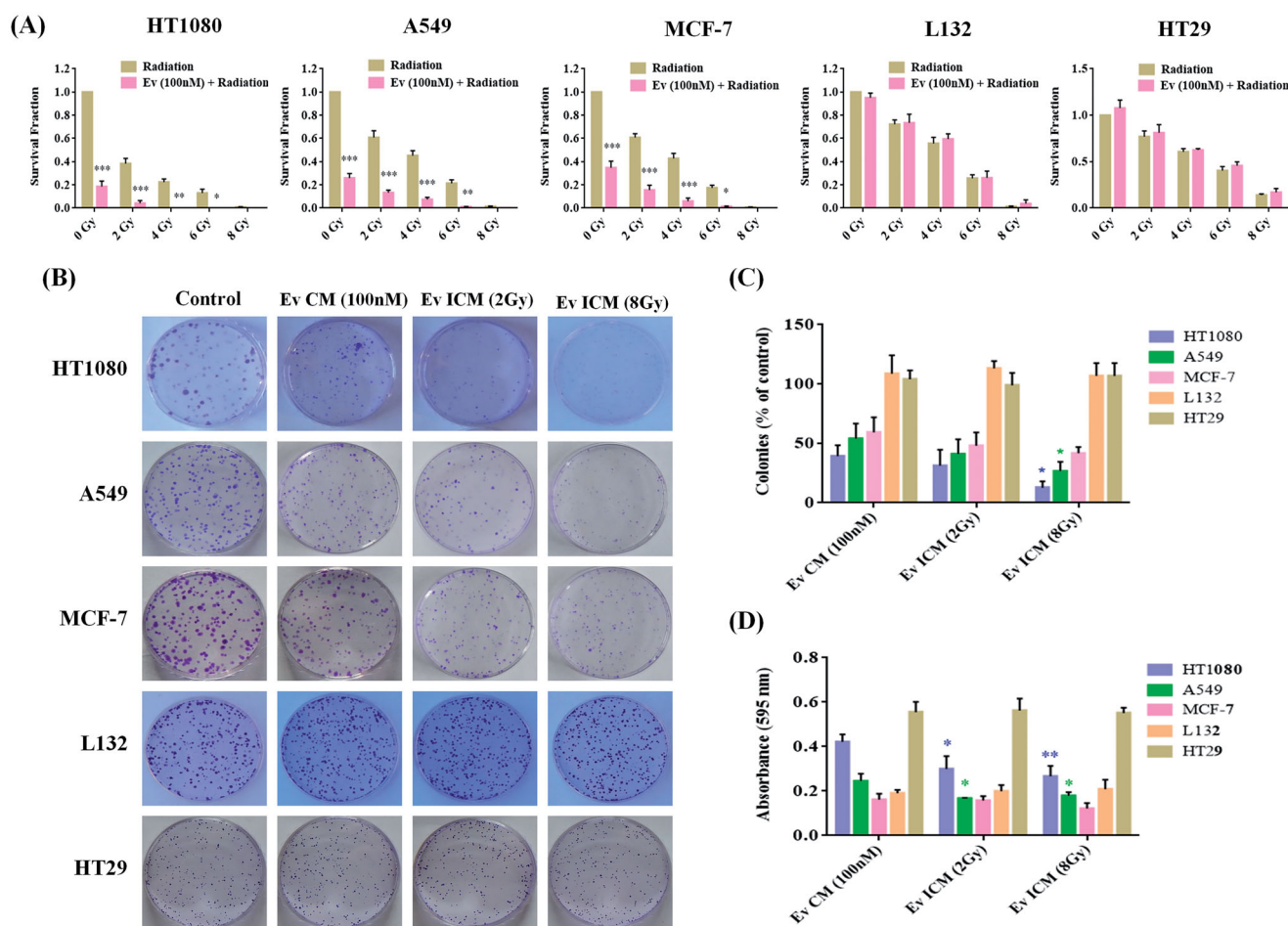


Figure 4. Direct and bystander effect of Ev + radiation treatments on cell survival. (A) Direct effect of 100 nM Ev + radiation using clonogenic assay. (B) Representative cell culture petri-dishes showing effect of Ev CM and Ev ICM treatment on colony formation. (C, D) Bar diagram showing comparative quantification of bystander effect induced by Ev CM and Ev ICM on naive HT1080, A549, MCF-7, L132 and HT29 cells using clonogenic and MTT assay, respectively. Data are means $\pm$ SD from three independent experiments, (* $p < .05$, ** $p < .01$, *** $p < .001$).

treatment with any of the conditioned medium (ICM or Ev ICM) (Figure 6(B) and Supplementary Figure 3(D)). These results indicated that activation/inhibition of the mTOR pathway plays an important role in conditioned media-induced proliferation or growth inhibition in bystander cells.

Discussion

RIBE is now a well-known biological phenomenon and can modulate (positively/negatively) the effective outcome of radiotherapy (Siva et al. 2015). Increased level of TNF- α , interleukins, cytokines, etc. in the serum of patients that have undergone radiotherapy is indicative of the role of bystander signaling in cancer therapy (Hong et al. 1999; Barker et al. 2018). In the present study, the media transfer method showed the induction of proliferative bystander effects. Out of five cell lines studied, conditioned media from two cell lines HT1080 and A549 induced proliferation in autologous bystander cells. Enhanced growth of bystander cells indicated the undesirable outcome of radiation treatment that may further lead to carcinogenesis or mutagenesis (Heeran et al. 2019). Proliferation in bystander cells has also been reported previously in rat liver epithelial (WB-F344) and normal fibroblast (GM637H) (Gerashchenko and

Howell 2003; Baskar et al. 2007). These proliferative bystander effects have been correlated with the presence of ROS, nitric oxide (NO), pro-inflammatory cytokines (TNF- α , IL-6), growth factors (PDGF-AA, PDGF-BB, TGF- α) and chemokines (fractalkine, MCP-1) in conditioned media (Baskar et al. 2007; Desai et al. 2013). Release of growth stimulatory, pro-inflammatory and pro-angiogenic factors may stimulate cell survival, rapid cancer growth, invasiveness and angiogenesis. Radiation-induced secretion of these factors is a strategy of tumor cells to maintain their tumor microenvironment for cell progression and metastasis (Landskron et al. 2014; Heeran et al. 2019). However, the inability of ICM from MCF-7, L132 and HT29 cells to induce bystander effects could be due to various reasons. The absence of bystander effect in MCF-7 cells has been correlated with the poor secretion of cytokines in conditioned media or due to their dependency on hormones rather than cytokines/growth factors (Desai et al. 2013). The absence of bystander effect in normal lung fibroblast (MRC-5) cells in comparison to lung carcinoma (QU-DB) cells has also been reported (Soleymanifard and Bahreyni 2012). Besides, inability of HSV-TK positive HT29 cells or ICM from 0.5 or 5 Gy γ -irradiated HT29 cells to induce bystander effects is also reported (McMasters et al. 1998; Yang et al.

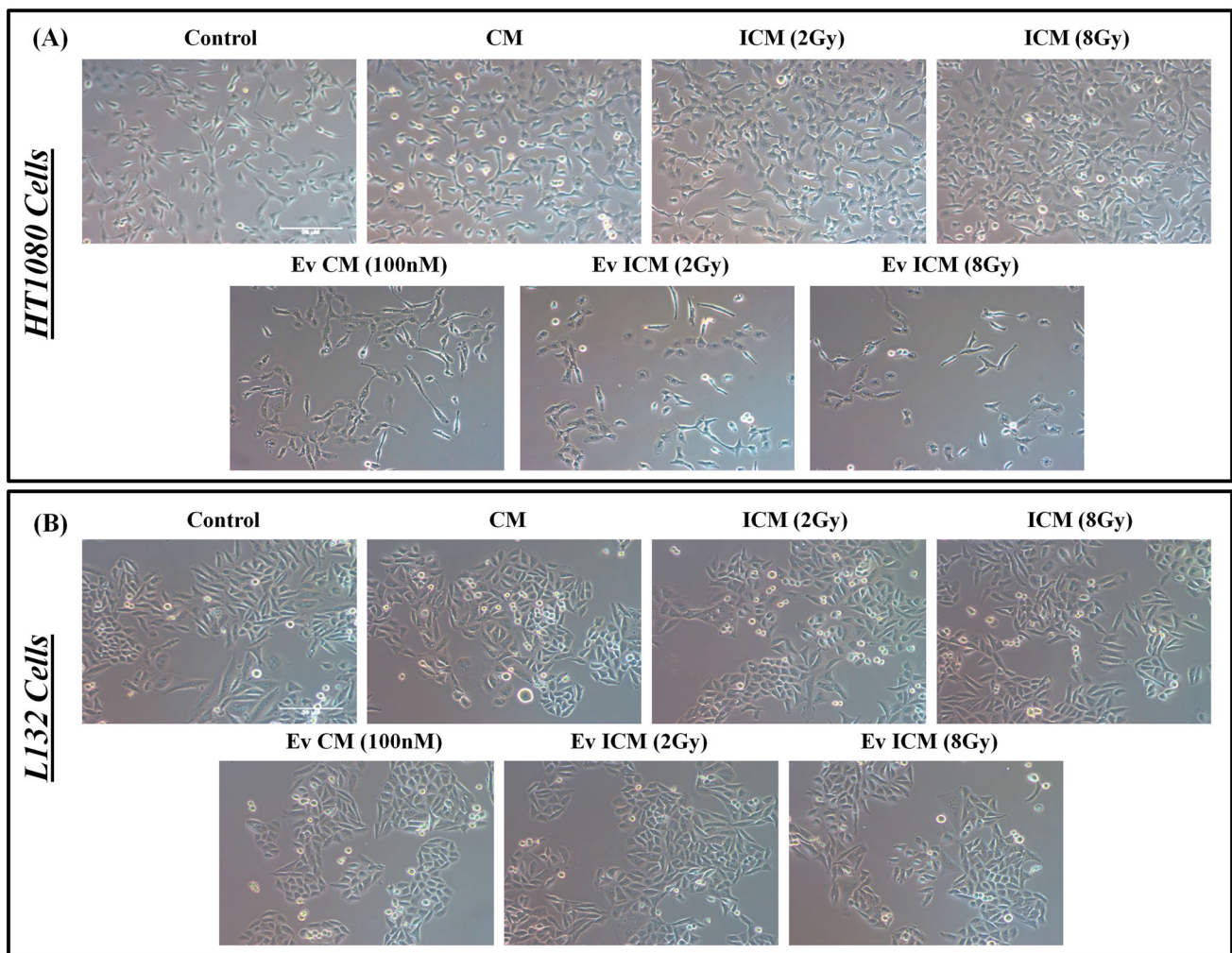


Figure 5. Representative photomicrographs showing effect of conditioned media obtained from different treatments on cell density and cell morphology of (A) HT1080 cells and (B) L132 cells.

1998; Ryan et al. 2008, 2009). The insensitivity of cells to media born signals were due to the absence of gap junction proteins, absence of receptors of signaling ligands, presence of radioresistance as well as the extent of DNA damage induced by radiation. So, these could be some of the factors responsible for the absence of bystander effects in the present study also.

Various signaling pathways like COX-2, MAPK, ERK, JNK, etc. also have a crucial role in the induction of bystander effects (Prise and O'Sullivan 2009; Wang et al. 2018). However, the role of the mTOR pathway in the modulation of RIBE is less explored (Li et al. 2017; Park et al. 2020; Verma and Tiku 2021). In the present study, irradiated HT1080 cells showed a significantly elevated expression of p-Akt, p-mTOR, and p-P70S6K proteins. mTOR pathway is reported to be associated with the production of pro-inflammatory cytokines (IL-6, IL-8, TNF- α , IFN- γ , and VEGF) (Soliman 2013; He et al. 2014). Thus, mTOR pathway activation in HT1080 cells and increased cell proliferation of ICM treated HT1080 cells could be due to the production and secretion of radiation-induced pro-inflammatory cytokines in the conditioned media.

Inhibition of mTOR by Ev increased the toxicity as well as radiosensitivity of HT1080, A549, and MCF-7 cells while

no toxic effect was observed in L132 and HT29 cells. Reduced cell survival in various cell lines after inhibition of the mTOR pathway is already reported (Tian et al. 2019). The nontoxic effect of mTOR inhibitors is also reported which might be either due to the intrinsic resistance of cells to inhibitors of the mTOR complex or due to the expression of the high level of Akt and mTOR proteins (Hassan et al. 2014). mTOR inhibition can offset the effect of radiation in normal, as well as stem cell lines as mTOR pathway is reported to have an effective role in DNA repair system and cell type-specific activity (Iglesias-Bartolome et al. 2012; Zheng et al. 2016). They can inhibit ROS and oxidative stress generated due to radiation via enhanced expression of MnSOD (mitochondrial superoxide dismutase). Further, the high endogenous expression of miR144 was also found to play an important role in providing resistance to mTOR inhibitors in HT29 cells (Iwaya et al. 2012).

Although Ev or Ev + radiation treatment decreased the expression of mTOR pathway proteins but unexpectedly an elevated expression of p-Akt could be due to the negative feedback loop of the Akt survival pathway that seems to attenuate Ev induced growth inhibitory/toxic effect in all cell lines. Previous studies also indicate increased phosphorylation of Akt after treatment with mTOR inhibitor in

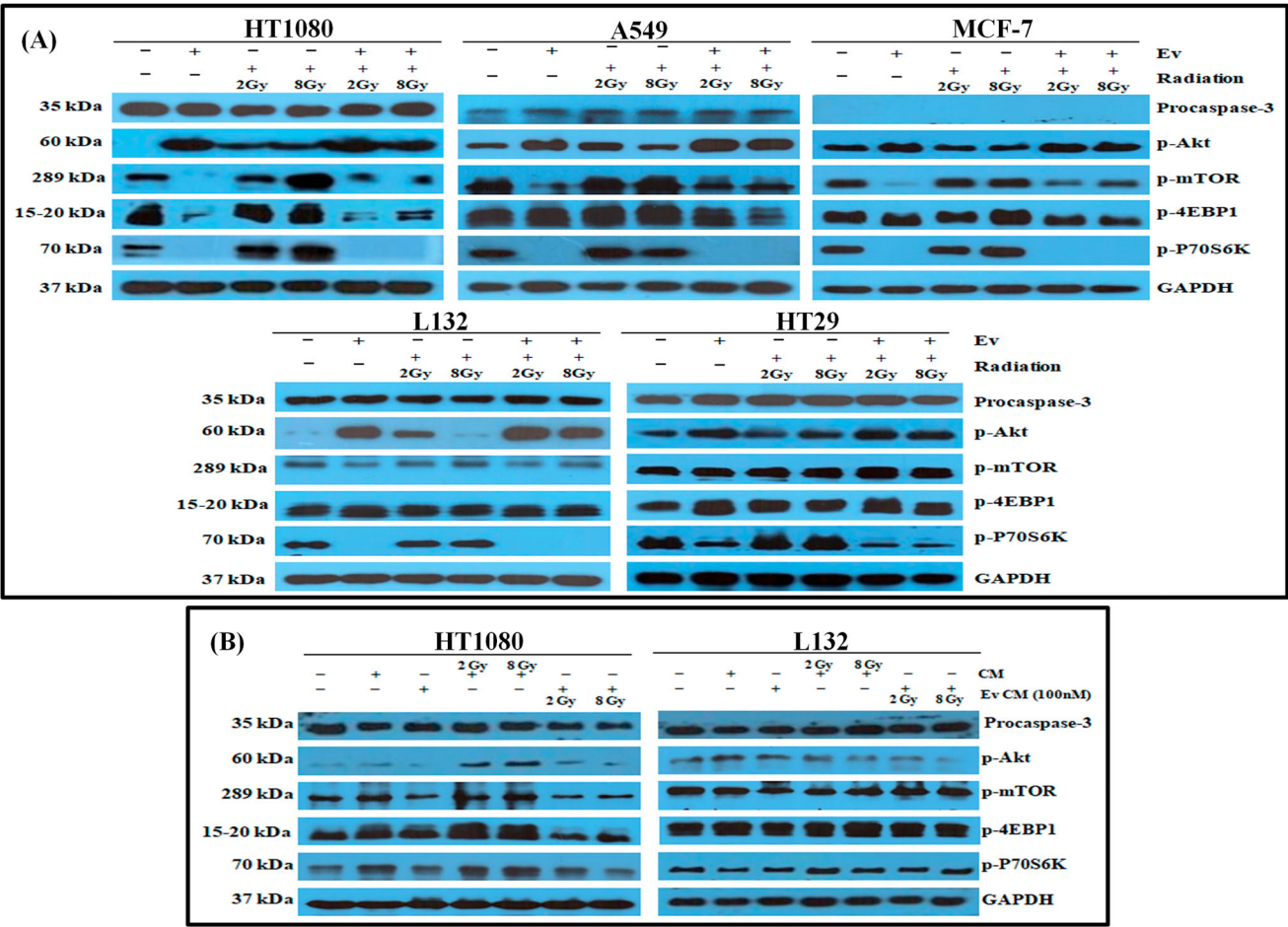


Figure 6. Effect of different treatments on the expression of mTOR pathway proteins. (A) Western blots indicating the expression of different proteins after treatment with Ev, radiation and Ev + radiation. (B) Western blots indicating the expression of different proteins in bystander HT1080 and L132 cells after treatment with different conditioned media (CM, ICM, Ev CM & Ev ICM). Data are representative blots of three independent experiments.

mammalian cell lines (Breuleux et al. 2009) as well as drosophila (Stocker et al. 2003). A significantly reduced plating efficiency and increased radiosensitivity in Ev + radiation treatment group might be due to the suppression of DNA double-strand repair mechanisms after mTOR inhibitor treatment. mTOR inhibitors are reported to impair the recruitment of Rad51 and BRCA1 proteins to DNA repair foci and thus radiosensitize the cells by disrupting the DNA repair pathway (Chen et al. 2011).

Repair deficient cells induced more toxic RIBEs in naive cells (Nagasawa and Little 2002; Nagasawa et al. 2003). Similarly, CM from mismatch repair (MMR) deficient Raji10 cells reduced the survival of bystander HPV-G cells (Mothersill et al. 2006). In the present study also, a significantly reduced cell survival in bystander HT1080, A549 and MCF-7 cells after Ev ICM treatment indicates the role of Ev induced radiosensitivity in the modulation of the bystander effect. Ev ICM from L132 and HT29 cells which were relatively radioresistant cell lines among the five cell lines tested, did not induce a significant change in cell survival of bystander cells further suggesting an important role of radiosensitivity in RIBE. In normal lung and lung cancer cell lines also, the presence or absence of bystander effects

positively correlated with the extent of DNA damage induced by radiation (Soleymanifard and Bahreyni 2012).

Besides radiosensitivity, Duan et al. reported a direct correlation between mTOR inhibition and reduced expression of pro-inflammatory cytokines (Duan et al. 2018). In the present study also, a reduced expression of mTOR pathway proteins in Ev + radiation treated cells and reduced cell survival in bystander HT1080, A549 and MCF-7 cells after Ev ICM treatment could be due to the low level or absence of pro-inflammatory factors (IL-1 β , IL-6, TNF- α , etc.) in Ev ICM.

Increased expression of p-Akt, p-mTOR, p-4EBP1 and p-P70S6K proteins after ICM treatment and reduced expression of these proteins after Ev ICM treatment in HT1080 cells resulting in proliferative versus toxic effects clearly suggests the role of these activated proteins and secretome thereof in the induction of bystander effects. A probable role of mTOR in RIBEs is given in Figure 7.

This study provides a new insight to the therapeutic approach (using bystander effect) based on modulation of radioresponse via mTOR inhibition, that is, from proliferative bystander effect to non-proliferative (toxic) bystander effect in cancer cell lines. Further to confirm and validate

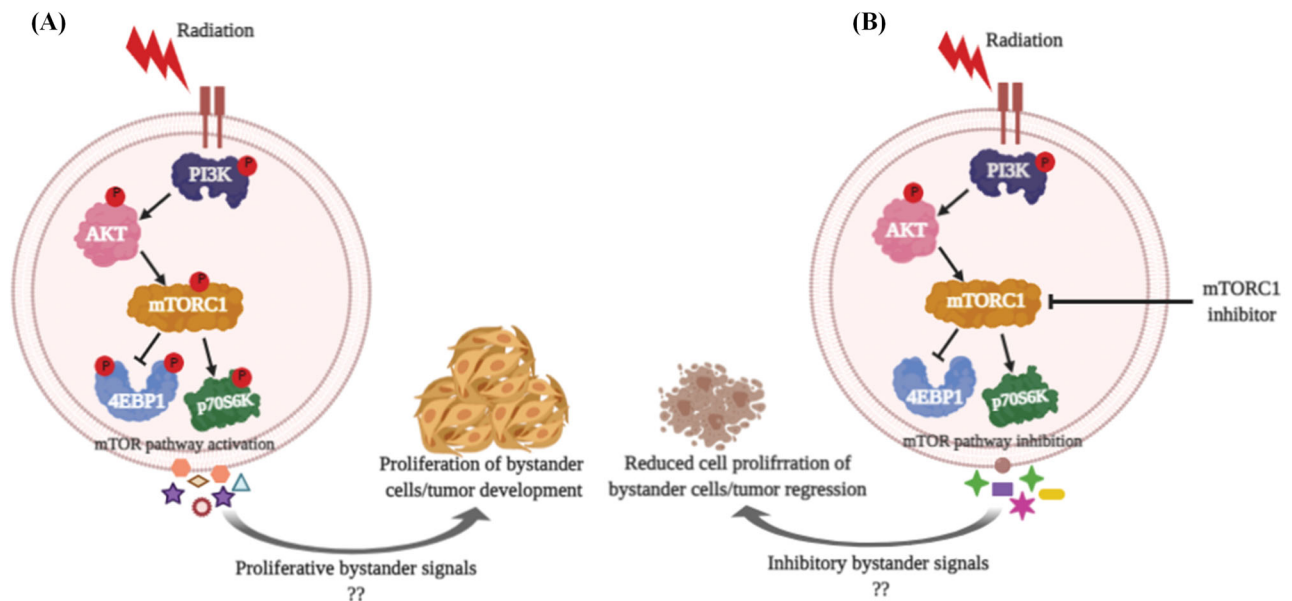


Figure 7. Proposed mechanism for induction of bystander effect. (A) Radiation induced activation of mTOR pathway in targeted cells could lead to tumor development in naïve cells by releasing proliferative signals. (B) Presence of mTORC1 inhibitor along with radiation can modulate the nature of bystander signals released by the targeted cells (from proliferative to inhibitory signals) which could lead to tumor regression in naïve cells.

the underlying mechanism, studies with a preclinical model need to be performed.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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